

SAUTRIK ROY

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EDUCATION

SRM Institute of Science and Technology (SRM IST)

B.Tech in Computer Science & Engineering (Core)

Kattankulathur, Chennai, India

Expected May 2029 (Batch 2025–2029)

- **CGPA:** 8.77 / 10.0 | **Coursework:** Data Structures & Algorithms, Object-Oriented Programming (C++), Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering.

TECHNICAL SKILLS

Programming Languages: C/C++, TypeScript, JavaScript (ES6+), Python, Java, SQL

Frontend: React, TanStack Start, HTML5, CSS3, Tailwind CSS, WebRTC, PWA, Context API

Backend & Cloud: Node.js, Express.js, REST APIs, Microsoft Azure, Supabase, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, Database Management Systems (DBMS)

DevOps, Testing & Tools: Git/GitHub, GitHub Actions, CI/CD, Docker, Linux, macOS, Postman, Vitest/Jest

CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, Software Engineering, System Design, Agile

EXPERIENCE & OPEN SOURCE

Microsoft Learn Student Ambassador (MLSA) — Campus Ambassador

Jan 2025 – Present

- Spearheaded 4+ technical workshops empowering 200+ campus developers in Azure Cloud architecture, Docker containerization fundamentals, and full-stack software development best practices.
- Mentored 50+ junior students in developer tooling and cloud fundamentals, promoting collaborative Git/GitHub workflows, code reviews, and modern software engineering standards.

GirlScript Summer of Code (GSSoC) — Open Source Contributor

2025

- Merged 8+ pull requests across distributed open-source repositories; resolved 5 frontend bugs and optimized component render cycles in React and TypeScript codebases.
- Collaborated with maintainers via Git/GitHub in an Agile setting, conducting peer code reviews, verifying PRs through GitHub Actions CI/CD workflows, and authoring unit tests using Vitest/Jest.

TECHNICAL PROJECTS

Loop Beyond Limits — Full-Stack Music Streaming Platform [► Live](#) | [GitHub](#)

TypeScript • React • TanStack Start • Supabase

- Architected an ad-free music streaming platform utilizing React, TanStack Start, and Supabase; implemented Server-Side Rendering (SSR) to cut initial page load time by ~40% by eliminating client-side render cycles.
- Engineered secure authentication flows, user input validation, and Row-Level Security (RLS) policies with real-time WebSocket state synchronization in PostgreSQL/Supabase to enable concurrent multi-user access without data conflicts.

Sentinel — AI Mental Health & Telemedicine Platform (SIH) [► Live](#) | [GitHub](#)

React • Node.js • Express.js • Supabase • WebRTC • PWA

- Developed an AI mental health platform for India's Uniformed Forces (Smart India Hackathon — Ministry of Home Affairs); engineered predictive scoring algorithms tracking 5 longitudinal signals (stress, sleep, fatigue, mood, energy) to screen for burnout.
- Built Node.js and Express.js REST APIs with Supabase backend, WebRTC video consultations, and an offline-first PWA caching records in IndexedDB—preserving 100% of queued offline assessments with sub-second local response time.

LegacyLens 2.0 — AI Natural Language to Database Query Interface [► Live](#) | [GitHub](#)

TypeScript • MySQL • MongoDB • NLP

- Built a web interface converting plain-English input into valid MySQL and MongoDB queries using NLP techniques, allowing users to query relational and document databases without writing SQL.
- Implemented parameterized queries and input validation for SQL injection prevention; benchmarked query generation against 10+ query patterns to verify accuracy across both relational and non-relational databases.

Quizzify — AI-Powered Competitive Assessment Platform [► Live](#) | [GitHub](#)

C++ • Google Gemini API • REST APIs

- Developed a high-speed C++ parsing backend to extract text from multi-page PDFs, integrating Google Gemini API via REST endpoints (tested via Postman) to automatically generate structured quiz questions in under 10 seconds.
- Applied OOP inheritance and polymorphism in C++ to implement modular Test Mode and Practice Mode architectures with shared base class logic, eliminating code duplication and streamlining mode extensibility.

HONORS & CERTIFICATIONS

• **Smart India Hackathon (SIH) 2025:** Shortlisted at National Level by Ministry of Home Affairs (MHA) for AI Welfare Monitoring System (SIH26186).

• **Certifications & Languages:** Microsoft Certified: Azure Fundamentals — In Progress • English (Professional), Bengali (Native), Hindi (Conversational), German (A1).